**NAME: THULASI S** **ROLL:NO:241901118**

# EXP 5: PERFORM WEB SERVER RECONNAISSANCE USING VARIOUS TOOLS

## AIM:

To identify the type, version, and configuration details of a target web server using reconnaissance tools such as Nmap, Nikto, and WhatWeb.

## THEORY:

Web server reconnaissance means collecting technical details about the web server software running on a target machine before testing it further. This includes:

- Web server software and version (Apache, Nginx, IIS, Tomcat)
- Operating system running underneath
- Open ports and running services
- HTTP response headers (which often reveal server software/version)
- Known default files, directories, or misconfigurations

This information is important because different web server versions have different known vulnerabilities — the more precisely we identify the server, the more precisely we can plan later testing steps (like vulnerability scanning in Experiment 9).

## TOOLS REQUIRED:

- Nmap (with -sV and NSE scripts)
- Nikto
- WhatWeb
- curl (to inspect HTTP headers directly)
- Target: Metasploitable2 (running Apache) or DVWA server

## PROCEDURE

Step 1: Basic Port and Service Scan with Nmap

Run a service/version detection scan against the target's IP:

```
nmap -sV -p 80,443,8080 <target-ip>
```

This shows which ports are open and attempts to identify the exact web server software and version running on each port.

Step 2: Run Nmap's HTTP Enumeration Scripts

Nmap includes built-in scripts (NSE) specifically for web server recon:

```
nmap -p80 --script=http-headers,http-title,http-server-header <target-ip>
```

This returns the server's response headers, page title, and the Server: header value directly.

Step 3: Inspect HTTP Headers Manually with curl

```
curl -I http://<target-ip>
```

The -I flag fetches only the HTTP headers. Look for the Server: header — it often reveals the exact web server name and version (e.g., Apache/2.2.8 (Ubuntu) PHP/5.2.4).

Step 4: Fingerprint the Web Technology Stack with WhatWeb

```
whatweb <target-ip>
```

WhatWeb identifies the CMS, JavaScript frameworks, analytics tools, and server software being used, all in one command.

Step 5: Run a Web Server Scan with Nikto

```
nikto -h http://<target-ip>
```

Nikto checks the web server for:

- Outdated server software
- Dangerous default files (like /phpinfo.php)
- Missing security headers
- Directories that should not be publicly accessible

This scan may take a few minutes — let it complete and review the list of findings it prints.

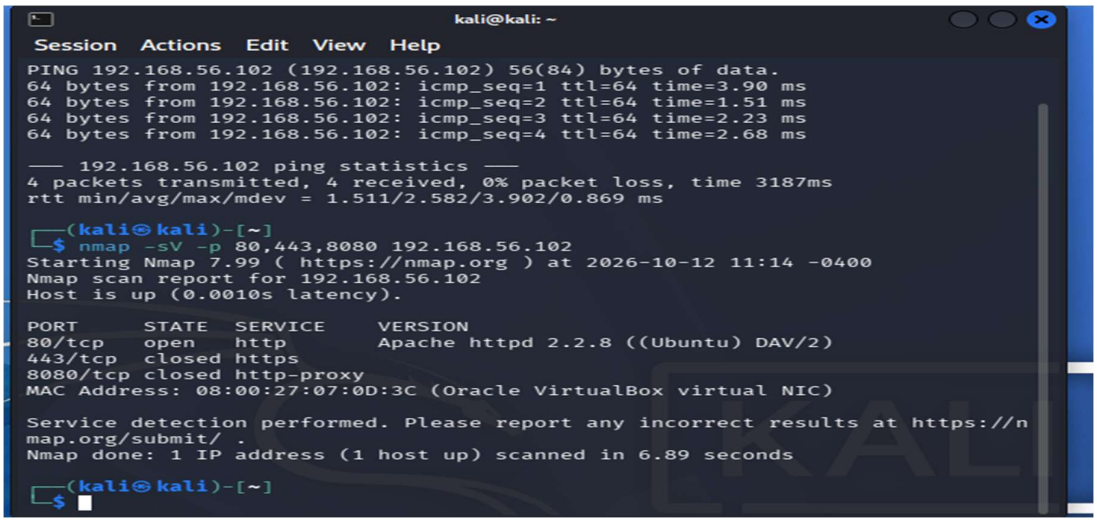
Step 6: Record the Server Fingerprint

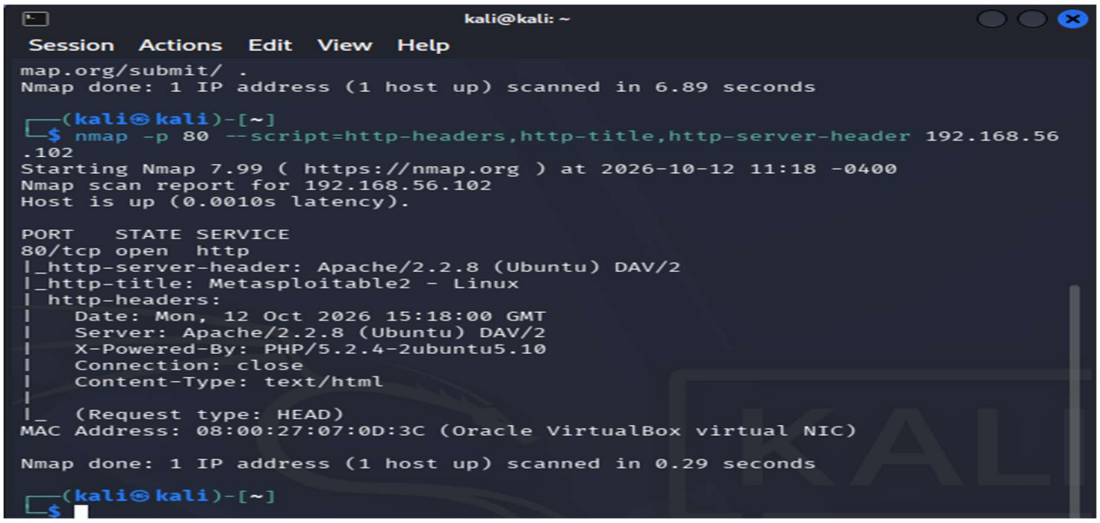
Fill in a summary table:

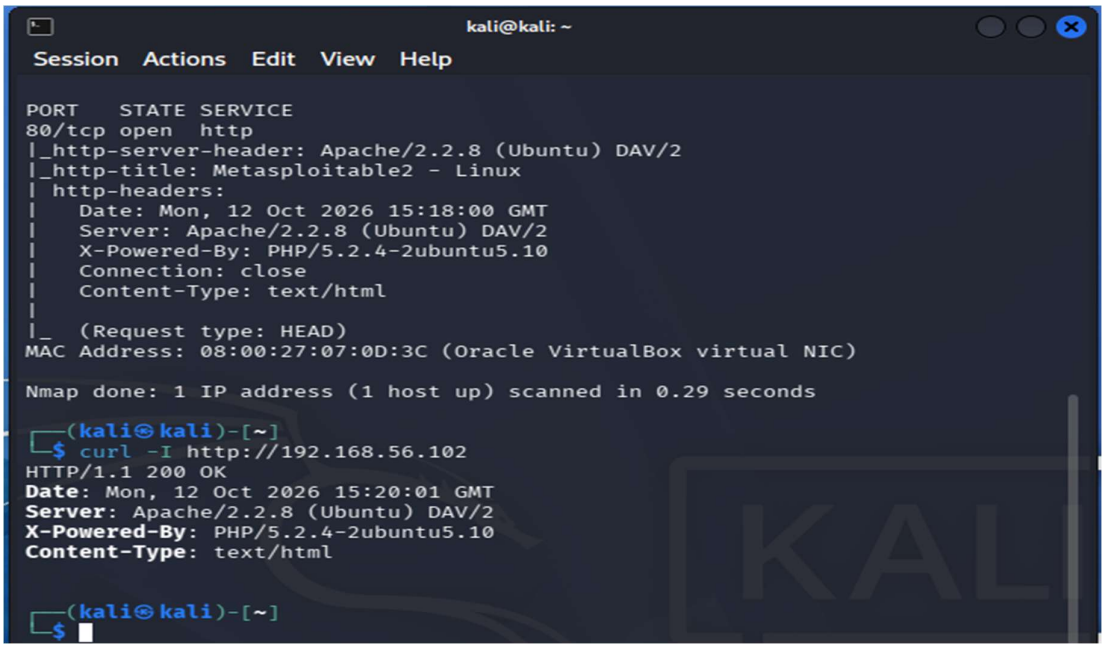
| Detail | Value Found |
|---|---|
| Web Server Software | Apache 2.2.8 |
| Operating System | Ubuntu (from headers/Nmap OS guess) |
| Open Ports | 80, 8080 |
| CMS/Framework | (if any found) |
| Security Issues Found (Nikto) | e.g., outdated PHP version |

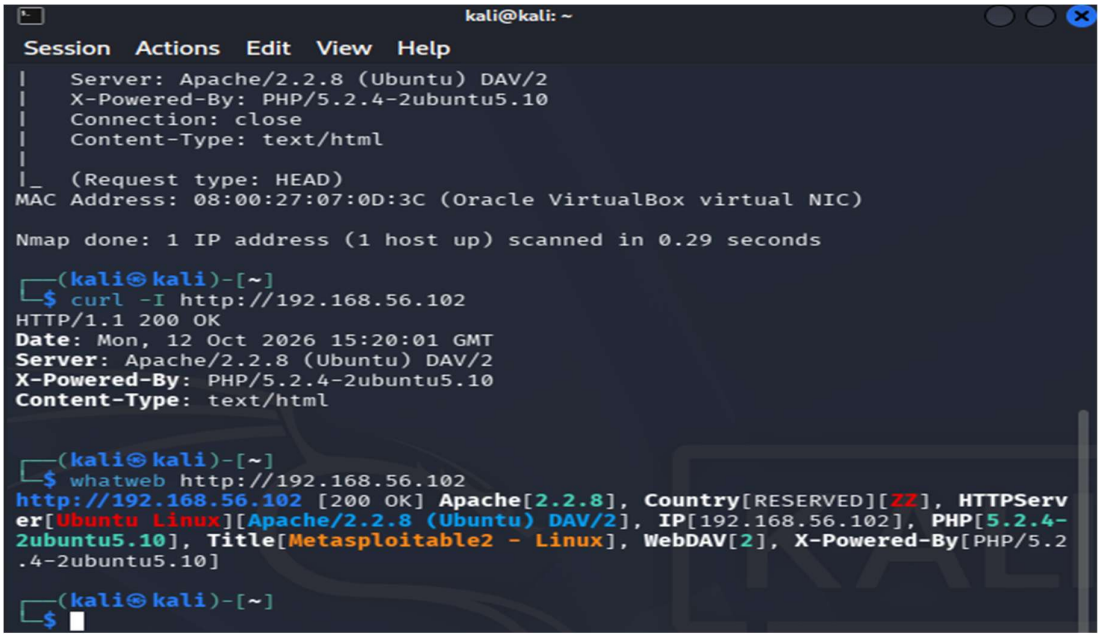
**OBSERVATION:**

| TEST PERFORMED | OBSERVATION |
|---|---|
| Network connectivity using ping | Target 192.168.56.102 was reachable. Four packets were received with 0% packet loss. |
| Nmap service/version detection | Port 80/tcp was open and running Apache HTTPD 2.2.8 (Ubuntu) DAV/2. Ports 443/tcp and 8080/tcp were closed. |
| Nmap HTTP scripts | Page title: Metasploitable2 - Linux. Server header: Apache/2.2.8 (Ubuntu) DAV/2. PHP header: PHP/5.2.4-2ubuntu5.10. |
| HTTP header inspection using curl | Server returned HTTP/1.1 200 OK. Apache and PHP response headers were observed; content type was text/html. |
| Web technology detection using WhatWeb | Detected Apache 2.2.8, Ubuntu Linux, PHP 5.2.4-2ubuntu5.10, page title Metasploitable2 - Linux, and WebDAV 2. |
| Web server scan using Nikto | Reported directory indexing, missing recommended security headers, outdated Apache/PHP versions, a phpinfo() page, a browsable /doc/ directory, and HTTP TRACE enabled. |




kali@kali: ~
Session Actions Edit View Help
map.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.89 seconds
┌──(kali㉿kali)-[~]
└─$ nmap -p 80 --script=http-headers,http-title,http-server-header 192.168.56
.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-12 11:18 -0400
Nmap scan report for 192.168.56.102
Host is up (0.0010s latency).
PORT STATE SERVICE
80/tcp open http
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_http-title: Metasploitable2 - Linux
| http-headers:
| Date: Mon, 12 Oct 2026 15:18:00 GMT
| Server: Apache/2.2.8 (Ubuntu) DAV/2
| X-Powered-By: PHP/5.2.4-2ubuntu5.10
| Connection: close
| Content-Type: text/html
|
|_ (Request type: HEAD)
MAC Address: 08:00:27:07:0D:3C (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.29 seconds
┌──(kali㉿kali)-[~]
└─$


kali@kali: ~
Session Actions Edit View Help
PORT STATE SERVICE
80/tcp open http
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_http-title: Metasploitable2 - Linux
| http-headers:
| Date: Mon, 12 Oct 2026 15:18:00 GMT
| Server: Apache/2.2.8 (Ubuntu) DAV/2
| X-Powered-By: PHP/5.2.4-2ubuntu5.10
| Connection: close
| Content-Type: text/html
|
|_ (Request type: HEAD)
MAC Address: 08:00:27:07:0D:3C (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.29 seconds
┌──(kali㉿kali)-[~]
└─$ curl -I http://192.168.56.102
HTTP/1.1 200 OK
Date: Mon, 12 Oct 2026 15:20:01 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Content-Type: text/html
┌──(kali㉿kali)-[~]
└─$







## RESULT:

Web server reconnaissance was carried out using Nmap, WhatWeb, Nikto, and curl in an authorised lab environment to examine open ports, service versions, HTTP headers, web technologies, and potential security issues.